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MPDS1

TP n°3: HIVE

Matière : Traitement du Big data avancé

1. Création de table :

Q1 :

```
[cloudera@quickstart ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.p
roperties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive>
    >
    >
```

Q2 :

```
    >
    > CREATE TABLE records (year STRING, temperature INT, quality INT) ROW FORMA
T DELIMITED FIELDS TERMINATED BY '\t';
OK
Time taken: 1.75 seconds
hive>
```

Q3 : Où sont stockées les données de cette table ?

Les données sont stockées sous HDFS '**/user/hive/warehouse/records'.**

Q4 :

```
hive>  LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/data-hive/sample-with-tab.t
xt'OVERWRITE INTO TABLE records;
Loading data to table default.records
Table default.records stats: [numFiles=1, numRows=0, totalSize=51, rawDataSize=0]
OK
Time taken: 1.041 seconds
hive>
```

Q5 :

- Afficher la liste des tables

```
hive> SHOW TABLES;
OK
records
Time taken: 0.27 seconds, Fetched: 1 row(s)
```

• Afficher la structure de la table records

```
hive> DESCRIBE FORMATTED records;
OK
# col_name              data_type               comment

year                    string
temperature             int
quality                 int

# Detailed Table Information
Database:               default
Owner:                  cloudera
CreateTime:             Sat Feb 24 03:17:36 PST 2024
LastAccessTime:         UNKNOWN
```

• Afficher tous les enregistrements de la table records

```
hive> SELECT * FROM records;
OK
1950    0       1
1950    22      1
1950    -11     1
1949    111     1
1949    78      1
Time taken: 0.473 seconds, Fetched: 5 row(s)
hive> 
```

Q6 :

```
hive> SELECT DISTINCT year FROM records;
Query ID = cloudera_20240224035151_45951fad-36cd-4dee-96d4-9da62d5828a3
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1707166712860_0025, Tracking URL = http://quickstart.cloudera
8088/proxy/application_1707166712860_0025/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1707166712860_0025
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2024-02-24 03:52:03,096 Stage-1 map = 0%,  reduce = 0%
2024-02-24 03:52:13,419 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.93 se
2024-02-24 03:52:25,083 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 4.65
```

Q7

```
hive> LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/data-hive/stations-fixed-wid
th.txt' OVERWRITE INTO TABLE stations;
Loading data to table default.stations
Table default.stations stats: [numFiles=1, numRows=0, totalSize=2253120, rawDataS
ize=0]
OK
Time taken: 0.383 seconds
hive> 
```

Q8 :

- Afficher tous les enregistrements des stations situées à PARIS
- Supprimer la table records

```
hive> SELECT * FROM stations WHERE name = 'PARIS';
OK
Time taken: 0.313 seconds
hive> DROP TABLE records;
OK
Time taken: 0.678 seconds
hive>
```

2. Création de table externe

2.1. Table externe avec tabulations

Q1 :

```
hive> CREATE EXTERNAL TABLE records (year STRING, temperature INT, quality INT)
    > ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t'
    > LOCATION '/user/cloudera/records/'
    > ;
OK
Time taken: 0.068 seconds
```

Q2 :

```
hive> LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/data-hive/sample-with-tab.txt' INT
O TABLE records;
Loading data to table default.records
Table default.records stats: [numFiles=1, totalSize=51]
OK
Time taken: 0.219 seconds
```

Q3 :

```
hive> DESCRIBE FORMATTED records;
OK
# col_name              data_type               comment

year                    string
temperature             int
quality                 int

# Detailed Table Information
Database:               default
Owner:                  cloudera
CreateTime:             Sat Feb 24 04:50:06 PST 2024
LastAccessTime:         UNKNOWN
Protect Mode:           None
Retention:              0
Location:               hdfs://quickstart.cloudera:8020/user/cloudera/records
Table Type:             EXTERNAL_TABLE
Table Parameters:
        COLUMN_STATS_ACCURATE   true
```

2.2. Table externe avec champs de taille fixe

Q1 :

```
hive> CREATE EXTERNAL TABLE records2 (station STRING, year STRING, temperature INT, quality
    > INT)
    > ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.RegexSerDe'
    > WITH SERDEPROPERTIES ("input.regex"=".{4}(.{6}).{5}(.{4}).{68}\\+?(-?\\d{4})(\\d).*")
    > LOCATION '/user/cloudera/records2';
OK
Time taken: 0.042 seconds
```

Q2 :

```
hive> LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/data-hive/sample.txt' INTO TABLE records
2;
Loading data to table default.records2
Table default.records2 stats: [numFiles=1, totalSize=1777168]
OK
Time taken: 0.276 seconds
```

Q3 :

• Sélectionner seulement 10 lignes (enregistrements) de la table records2

```
hive> SELECT * FROM records2 LIMIT 10;
OK
029070  1901    -78     1
029070  1901    -72     1
029070  1901    -94     1
029070  1901    -61     1
029070  1901    -56     1
029070  1901    -28     1
029070  1901    -67     1
029070  1901    -33     1
029070  1901    -28     1
029070  1901    -33     1
Time taken: 0.044 seconds, Fetched: 10 row(s)
```

• Sélectionner de la table records2, les enregistrements ayant la qualité 9

```
hive> SELECT * FROM records2 WHERE quality = 9;
OK
029810  1901    9999    9
Time taken: 0.038 seconds, Fetched: 1 row(s)
hive>
```

3. Agrégats :

Q1 :

```
hive> SELECT year, MAX(temperature)
    > FROM records2
    > WHERE temperature != 9999
    > AND (quality=0 OR quality=1 OR quality=4 OR quality=5 OR quality=9)
    > GROUP BY year;
Query ID = cloudera_20240224050707_83b9e1cf-51cb-4004-a4bd-6c290195b724
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1707166712860_0026, Tracking URL = http://quickstart.cloudera:8(
pplication 1707166712860 0026/
```

```
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 4.14 sec   HDFS
18 SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 140 msec
OK
1901    317
1902    244
Time taken: 40.188 seconds, Fetched: 2 row(s)
```

Q2 :

```
hive> SELECT year, AVG(temperature)
    > FROM records2
    > WHERE temperature != 9999 AND (quality=0 OR quality=1 OR quality=4 OR quality=5 OR qual
ity=9)
    > GROUP BY year;
Query ID = cloudera_20240224050909_2d5358ba-8c5e-440d-bb62-4016cc4c5fdc
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
```

```
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 4.25 sec   HDFS Read: 1787324 HDFS
45 SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 250 msec
OK
1901    46.698507007922
1902    21.659558263518658
Time taken: 40.263 seconds, Fetched: 2 row(s)
```

4. Requêtes imbriquées

Q1 :

```
hive> SELECT AVG(max_temperature)
    > FROM (
    >  SELECT station, year, MAX(temperature) AS max_temperature
    >  FROM records2
    >  WHERE temperature != 9999 AND (quality=0 OR quality=1 OR quality=4 OR quality=5 OR qua
lity=9)
    >  GROUP BY station, year
    > ) mt;
Query ID = cloudera_20240224051414_8266c807-de91-4372-bc52-b5571ea77f06
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
```

```
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 4.13 sec
127 SUCCESS
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 3.05 sec
UCCESS
Total MapReduce CPU Time Spent: 7 seconds 180 msec
OK
246.0
Time taken: 78.189 seconds, Fetched: 1 row(s)
```

Q2 :

```
hive> SELECT FLOOR(avg_temperature/50), COUNT(station)
    > FROM (
    >  SELECT station, AVG(temperature) AS avg_temperature
    >  FROM records2
    >  WHERE temperature != 9999
    >  AND (quality=0 OR quality=1 OR quality=4 OR quality=5 OR quality=9)
    >  GROUP BY station )
    > mt GROUP BY FLOOR(avg_temperature/50);
Query ID = cloudera_20240224052323_f85a7534-318f-4c66-809f-60ba0ba13cf6
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
```

```
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 5.12 sec   HDFS Read: 1787384 HDF
134 SUCCESS
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 2.58 sec   HDFS Read: 4989 HDFS W
UCCESS
Total MapReduce CPU Time Spent: 7 seconds 700 msec
OK
0       5
1       1
Time taken: 81.707 seconds, Fetched: 2 row(s)
```

5. Jointures

Q1 :

```
hive> SELECT DISTINCT name
    > FROM stations
    > JOIN records2 ON (records2.station = stations.usaf)
    > WHERE year = '1901' OR year = '1902';
Query ID = cloudera_20240224052525_6c529c0d-228e-456e-9f02-c6cb8f749973
Total jobs = 1
Execution log at: /tmp/cloudera/cloudera_20240224052525_6c529c0d-228e-456e-9f02-c6cb8f7499
log
2024-02-24 05:25:23     Starting to launch local task to process map join;      maximum mem
y = 1013645312
2024-02-24 05:25:27     Dump the side-table for tag: 1 with group count: 6 into file: file
mp/cloudera/3bc7f82c-5419-4e14-8a27-ceb4a872913b/hive_2024-02-24_05-25-14_802_640269152716
0428-1/-local-10004/HashTable-Stage-2/MapJoin-mapfile01--.hashtable
2024-02-24 05:25:27     Uploaded 1 File to: file:/tmp/cloudera/3bc7f82c-5419-4e14-8a27-ceb
```

```
MapReduce Jobs Launched:
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 4.11 sec
150 SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 110 msec
OK

TURKU
ULKOKALLA
UTO
VYARTSILYA
```

Q2 :

- Calcul de la température moyenne de l'ensemble des stations météo UTO et TURKU.

```
hive> SELECT AVG(temperature) AS average_temperature
    > FROM records2
    > JOIN stations ON records2.station = stations.usaf
    > WHERE stations.name IN ('UTO', 'TURKU');
Query ID = cloudera_20240224054040_fdbad017-38b3-4722-aeaf-9985ea2b0ca2
Total jobs = 1
Execution log at: /tmp/cloudera/cloudera_20240224054040_fdbad017-38b3-4722-a
log
2024-02-24 05:40:58     Starting to launch local task to process map join;
y = 1013645312
```

- Calcul de la température moyenne de la station météo UTO, et de celle de TURKU.

```
hive> SELECT AVG(temperature) AS average_temperature_uto
    > FROM records2
    > JOIN stations ON records2.station = stations.usaf
    > WHERE stations.name = 'UTO';
Query ID = cloudera_20240224053333_f6945d51-cc95-4260-a248-c6b1edd551f9
Total jobs = 1
Execution log at: /tmp/cloudera/cloudera_20240224053333_f6945d51-cc95-4260-a248-
log
2024-02-24 05:33:53     Starting to launch local task to process map join;
y = 1013645312
```

```
hive> SELECT AVG(temperature) AS average_temperature_turku
    > FROM records2
    > JOIN stations ON records2.station = stations.usaf
    > WHERE stations.name = 'TURKU';
Query ID = cloudera_20240224053434_b6008ef7-ff93-4bfc-8f9c-0eb2cfc734a4
Total jobs = 1
Execution log at: /tmp/cloudera/cloudera_20240224053434_b6008ef7-ff93-4bfc-8f9c-0eb2cfc734a
log
2024-02-24 05:35:04     Starting to launch local task to process map join;      maximum mem
y = 1013645312
2024-02-24 05:35:08     Dump the side-table for tag: 0 with group count: 6 into file: file:
```

6. Tri

Q1 :

```
hive> SELECT *
    > FROM records2
    > WHERE temperature != 9999 AND (quality = 0 OR quality = 1 OR quality = 4 OR quality =
 OR quality = 9)
    > SORT BY year ASC, temperature DESC;
Query ID = cloudera_20240224054343_c8e7c14f-254a-4bce-9ffd-a0cfb749b5a7
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
```

```
029500  1902    -117    1
029070  1902    -117    1
029600  1902    -117    1
029070  1902    -117    1
029500  1902    -117    1
029500  1902    -117    1
029600  1902    -117    1
029500  1902    -117    1
029500  1902    -117    1
029600  1902    -117    1
029500  1902    -117    1
227070  1902    -117    1
227070  1902    -117    1
227070  1902    -117    1
227070  1902    -117    1
227070  1902    -117    1
029070  1902    -117    1
029070  1902    -117    1
029070  1902    -117    1
029070  1902    -117    1
029720  1902    -117    1
029070  1902    -117    1
029070  1902    -117    1
029600  1902    -117    1
```